
Liar, Liar: Beyond Vocabulary Suppression

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github.com/aryan-cs/liar-liar

1 Related work

Activation steering. Representation engineering adds contrastively constructed vectors to the residual stream to modulate high-level behavior (Zou et al., 2023); contrastive activation addition (Rimsky et al., 2024) and inference-time intervention (Li et al., 2023) are the canonical honesty-adjacent instantiations, and Arditi et al. (2024) show refusal is mediated by a single direction that can be removed from every writing matrix. Tan et al. (2024) document that steering vectors often generalize poorly out of distribution; our depth statistic offers a mechanism-level account of when that should happen.

Concept erasure. LEACE (Belrose et al., 2023) gives the minimum-norm affine map that removes linear decodability of a concept, following iterative nullspace projection (Ravfogel et al., 2020) and rank-constrained adversarial erasure (Ravfogel et al., 2022). Our projection is structurally the LEACE construction with two changes: the erased subspace is read directly off the model’s effective unembedding rather than learned from labels, and the constraint is zero direct logit contribution on a token set rather than zero probe accuracy.

Unembedding geometry. Park et al. (2024) formalize the duality between context-side and unembedding-side concept directions under a causal inner product; our rank-one variant projects against the unembedding-side honesty direction, and our Mahalanobis-weighted variant recovers their inner product. Marks and Tegmark (2024) construct truth directions by mass-mean differences, which we use both as an alternative steering vector and as the readout-side direction of the rank-one test. The direct logit-attribution path through the residual stream is the lens of Elhage et al. (2021) and nostalgebraist (2020).

Closest precedents. Three works overlap most directly. Venkatesh and Kurapath (2026) prove steering vectors are non-identifiable up to perturbations in the null space of the full activation-to-logit Jacobian; our subspace is the readout-restricted version, and where they establish ambiguity we measure what the ambiguity carries. Nadaf (2026) show function vectors steer behavior the logit lens cannot decode, establishing the off-readout channel in the in-context-learning setting. The unembedding-steering benchmark of van Deventer (2024) implements W_U -orthogonal steering for sentiment on Gemma-2-9B. None applies the construction to honesty steering, none corrects for the RMSNorm Jacobian, and none reports a quantitative depth decomposition with certificates; those are the contributions here.

2 Method

2.1 Setup and notation

Let $\mathcal{M}$ be a decoder-only transformer with L layers, residual width d , and vocabulary size V . The residual stream $h_i^{(\ell)} \in \mathbb{R}^d$ is updated additively by each block, and the logits at the final position are

$$\ell(h_n^{(L)}) = W_U \cdot \text{RMSNorm}_\gamma(h_n^{(L)}), \quad \text{RMSNorm}_\gamma(z) = \gamma \odot \frac{z}{\sqrt{d^{-1}\|z\|_2^2 + \varepsilon}}, \quad (1)$$

with $W_U \in \mathbb{R}^{V \times d}$ the unembedding and $\gamma \in \mathbb{R}^d$ a learned gain (Zhang and Sennrich, 2019). A steering intervention adds a vector αv_{dec} to the residual stream at a middle layer ℓ^* at every position. We use the contrastive mean-difference construction (Rimsky et al., 2024; Zou et al., 2023): given prompts rendered under an honest and a deceptive system prompt, v_{dec} is the difference of mean final-token residuals at ℓ^* .

Effective unembedding. For a perturbation δz at the readout point z , the first-order logit response is governed not by W_U but by the composition with the RMSNorm Jacobian,

$$\widetilde{W}_U(z) := W_U J_\gamma(z), \quad J_\gamma(z) = \frac{1}{\sigma(z)} \text{diag}(\gamma) \left(I_d - \frac{zz^\top}{d\sigma(z)^2} \right), \quad (2)$$

where $\sigma(z) = \sqrt{d^{-1}\|z\|_2^2 + \varepsilon}$. Projecting against raw W_U rows, as in prior unembedding-orthogonal constructions, leaves a residual direct effect through the gain and the rank-one term. We average J_γ over a calibration set of readout points and write $\widetilde{W}_U^*$ for the resulting matrix.

2.2 Token-conditional orthogonalization

A global version of the test — a perturbation invisible to *every* token — is impossible: for every model we consider, $V > d$ and W_U has full column rank, so $\ker(W_U) = \{0\}$ (Proposition 3.1 of the supplementary proof). The construction must restrict the readout to a chosen token set.

Fix a deception-coded token set $T \subset \{1, \dots, V\}$ with $k = |T| \ll d$, and let $A = \widetilde{W}_U^*[T, :] \in \mathbb{R}^{k \times d}$ be the corresponding rows. With A^+ the Moore–Penrose pseudoinverse,

$$P_T^\perp = I_d - A^+ A, \quad v^\perp = P_T^\perp v_{\text{dec}}, \quad v^\parallel = v_{\text{dec}} - v^\perp. \quad (3)$$

By construction $A v^\perp = 0$: the projected vector has exactly zero first-order logit contribution at every token in T . Among all vectors satisfying that constraint, $v^\perp$ is the one closest to v_{dec} in Euclidean norm (the LEACE-style minimum-norm characterization; Theorem 8.1 of the supplementary proof, after Belrose et al., 2023).

2.3 Direct and indirect paths

Writing $M^{(\ell^* \rightarrow L)}$ for the summed Jacobian of all attention and MLP contributions downstream of ℓ^* , the first-order logit response to an injected v decomposes as

$$\Delta \ell = \underbrace{\widetilde{W}_U^* v}_{\text{direct}} + \underbrace{\widetilde{W}_U^* M^{(\ell^* \rightarrow L)} v}_{\text{indirect}} + O(\|v\|^2). \quad (4)$$

For $v = v^\perp$ the direct term vanishes on T identically, so any change in the T -restricted logits — and, to the extent that behavior on a deception benchmark is mediated by T , any change in measured behavior — must travel through downstream computation (Theorem 6.1 of the supplementary proof).

2.4 The depth statistic

Let $\beta(v)$ be a benchmark score under intervention v and $\Delta(v) = \beta(v) - \beta(0)$. The depth statistic is

$$\rho(v_{\text{dec}}, T) = \frac{\Delta(v^\perp)}{\Delta(v_{\text{dec}})}. \quad (5)$$

Under a pure vocabulary-suppression account, $\rho \approx 0$ once T exhausts the readout-aligned mass of v_{dec} ; under a pure upstream-concept account, $\rho \approx 1$. We report ρ with paired bootstrap confidence intervals, its stability σ_T across independent constructions of T , and a per-prompt analogue ρ_η defined on the honest-shift

$$\eta(v) = [\mu_{T^+}(v) - \mu_{T^-}(v)] - [\mu_{T^+}(0) - \mu_{T^-}(0)], \quad (6)$$

where $\mu_{T^\pm}$ is the mean logit over honest-coded (T^+) and deceptive-coded (T^-) tokens at the answer position.

Token-set constructions. We instantiate T three ways. *Curated*: a fixed lexicon of 32 honest and 32 deceptive lemmas expanded to single-token surface variants. *Statistical*: the top-32 tokens per side by mean first-position logit shift between honest- and deceptive-prompted contexts on held-out prompts. *Aligned- k* : the top- k tokens by $|\widetilde{W}_U^* v_{\text{dec}}|$ — the k tokens the steering vector moves most through the direct path. The aligned construction is the strongest form of the test: it removes exactly the direct-readout content of v_{dec} itself, and sweeping k traces how the surviving effect $\rho(k)$ decays as more of the direct path is excised.

Controls. Three controls separate the unembedding subspace from generic perturbation. (i) *Random projection*: project out a random k -dimensional subspace (three seeds), matching the dimension count of the aligned-64 set. (ii) *Norm matching*: rescale $v^\perp$ to $\|v_{\text{dec}}\|$, separating direction from magnitude. (iii) *Parallel injection*: inject $v^\parallel$ alone, the readout-aligned rank- $\leq k$ component; under the suppression account this component should carry the behavioral effect.

3 Experimental setup

Model. All experiments use Llama-3-8B-Instruct (Grattafiori et al., 2024) ($L = 32$, $d = 4096$, $V = 128,256$, RMSNorm) in bfloat16. Projections and certificates are computed in float64.

Data and splits. TruthfulQA (Lin et al., 2022) provides the behavioral benchmark in its multiple-choice form (817 questions). A seeded shuffle assigns 120 questions to a validation slice used only for the layer/strength sweep, 200 to construct mass-mean statements, and the remaining 497 to the held-out test slice on which every headline number is computed. Steering vectors are built from 256 Alpaca instructions (Taori et al., 2023) rendered under an honest and a deceptive system prompt; a further 128 disjoint instructions feed the statistical token set. Steering-vector data and evaluation data share no prompts.

Steering vectors. The CAA vector is the mean difference of final-token residuals between honest- and deceptive-prompted renderings, computed per layer. The intervention layer and strength are selected on the validation slice over $\ell \in \{10, 12, 14, 16\}$, $\alpha \in \{4, 8, 12\}$ by MC2 improvement, giving $\ell^* = 10$, $\alpha^* = 8$. A Marks–Tegmark mass-mean vector (Marks and Tegmark, 2024) built from true/false statement contrasts at ℓ^* provides a second construction.

Scoring. TruthfulQA MC follows the original protocol: the total log-probability of each answer choice under the prompt Q: {question}\nA:, with MC1 the accuracy of the argmax choice and MC2 the normalized probability mass on true choices. The intervention is applied at every token position. At the answer position we additionally record the logits of every token in the union of all

token sets, from which the honest-shift η and the spillover readouts are computed without further forward passes.

Conditions. The headline matrix evaluates: baseline; v_{dec} ; $v^\perp$ for aligned- k with $k \in \{16, 64, 256, 1024\}$; $v^\perp$ for the curated and statistical sets; the parallel component $v^\parallel$ (aligned-64); the norm-matched $v^\perp$ (aligned-64); random 64-dimensional projections (three seeds); and the mass-mean vector. All conditions share ℓ^* and α^* .

Out-of-distribution probe. The first 150 test questions are paraphrased by the unsteered model (greedy decoding, meaning-preserving instruction) and rescored under baseline, v_{dec} , and $v^\perp$ (aligned-64), giving an OOD depth ratio ρ_{OOD} . The spillover readout evaluates η on stem-disjoint synonym sets that were never projected out.

Reproducibility. The pipeline is four checkpointed stages with fixed seeds; all code, token lists, and the formal proof are in the repository. The zero-direct-effect certificates $\max_t |(Av^\perp)_t|$ are reported alongside the results.

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